Decoding the Beige Book: LLM-Powered Sentiment Analysis for Real-Time Recession Forecasting

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Abstract

In this research, we apply large language models (LLMs) to systematically analyze the sentiment contained in the Federal Reserve's Beige Book, a key qualitative resource to evaluate U.S. macroeconomic conditions. Leveraging both general-purpose and finance-specific LLMs, we developed an automated pipeline to extract topic-level sentiment scores from over five decades of Beige Book reports. We compare national and district-level sentiment signals, benchmark multiple LLMs, and quantify the predictive power of different economic topics for recession risk across various forecast horizons, both in-sample and out-of-sample. Sentiments generated by multiple LLM models are able to confidently signal for recession nowcasting, with peak performance at the contemporaneous horizon (F1 scores up to 0.89). By combining LLMs and structured sentiment analysis, we see the potential for AI-powered sentiment analysis to supplement traditional macroeconomic indicators in predicting recession risks.

Keywords

Natural Language Processing, Recession Prediction, Large Language Models, Federal Reserve Beige Book, Nowcasting, Sentiment Analysis

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1 Introduction and Motivation

The Federal Reserve's Beige Book, published eight times a year approximately two weeks before each Federal Open Market Committee (FOMC) meeting, is a cornerstone of U.S. economic analysis. Since 1970, it has provided a unique real-time assessment of economic conditions by compiling qualitative and anecdotal information from business contacts, economists, and market experts across the 12 Federal Reserve Districts. Each publication includes detailed reports from the individual districts and a national summary that synthesizes these regional perspectives. This structure makes the Beige Book an invaluable resource for understanding the nuanced economic landscape that informs monetary policy decisions.

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Given its timeliness and rich narrative content, the Beige Book has long been a subject of economic research. However, its analytical potential has been constrained by a significant structural challenge: the lack of consistent topic categorization over its 50+ year history. Section headings, content, and structure vary significantly across time and between different district reports. The Beige Book before 2018 lacked standardized, clear topic labels, which made it exceedingly difficult to conduct systematic, long-term analysis of specific economic sectors (e.g., manufacturing, labor markets), undermining both the explanatory and causal value of any sentiment-based research on the Beige Book data. This paper addresses these gaps by harnessing modern large language models (LLMs) to systematically quantify and analyze the sentiment expressed in the Beige Book. These innovations allow us to move far beyond the traditional binary rating of each report, enabling us to answer a far more nuanced question: which specific sectors are truly driving shifts in overall economic sentiment? Our analysis reveals that sentiment in "Employment and Wages", "Manufacturing", and "Real Estate and Construction" are consistently the most significant real-time predictors of U.S. recessions. Our primary contributions are: 1) We create the first structurally uniform, topic-labeled text dataset from the entire Beige Book corpus, spanning 1970 to the present; 2) We develop highly granular, topic-specific sentiment indices for district-level and national-level data to predict recessions; 3) We conduct a rigorous comparison of five leading LLMs, including both closed and open-source models, on the task of financial text sentiment analysis; 4) We systematically compare the predictive power of national versus district-level sentiment and show that the district-level sentiment is better at recession nowcasting than national-level sentiment; and 5) We construct a real-time recession probability index using Beige Book data based on our research.

2 Related Work

The application of large language models (LLMs) to solve domain-specific problems in the field of finance started with the equity side. Lopez-Lira and Tang [15] used LLMs to forecast stock returns. Recent work by Kim et al. [14] suggested that prompt engineering, especially chain-of-thought, has proven useful in similar tasks.

In macroeconomics, Hansen and Kazinnik [11] evaluated how GPT models deciphered Fedspeak, the complex and nuanced language used by the Federal Reserve to communicate monetary policy decisions. Furthermore, it demonstrated the capabilities of LLMs in both zero-shot and few-shot learning settings, showing promising results in comprehending Fedspeak without domain-specific training and additional labeled data. This could have implications for applying LLMs to other fields with specialized language and complex analysis.

Recent work by Cleveland Fed researchers [6] constructed quantitative sentiment estimates using FinBERT from the Beige Book, showing that regionally disaggregated sentiment provides valuable predictive signals for U.S. recessions. One limitation mentioned by the authors is the lack of topic-level research on this data. As a source of financial language, some research has confirmed the Beige Book's predictive power using a Bag of Words approach, showing that recent pessimism has a greater impact on GDP predictions than optimism, although it lacked the ability to analyze language in context. [1] Others, such as Sadique et al. [19], demonstrated that the Beige Book is a strong predictor. Their results showed that certain negative words declined when a positive production trend began. On the other hand, a decreasing frequency of 'increase' words predicted upcoming unemployment phases. However, other findings suggest that the Beige Book does not necessarily improve the forecasting capabilities of other hard predictors released earlier in the same reporting cycle. One study showed similar results with or without the Beige Book, highlighting again the limitation of lacking contextual understanding to fully capture economic activity. [3]

From a different perspective, Saltzman and Yung [20] used the Beige Book to measure different types of uncertainty. Using Principal Component Analysis, they classified uncertainty and found that the 'Business and Economics' category was associated with a decline in GDP, whereas the 'Politics and Government' category didn't show statistical significance. The Beige Book has also been studied to determine its usefulness for asset allocation decisions [18], to predict yield spreads [7] and its influence on investment behavior in the Foreign Exchange Market [2] utilizing the FinBERT model.

The FinBERT model was designed for the finance domain; it outperformed other approaches and models, achieving significantly higher accuracy [12]and superior performance in financial sentiment analysis [23]. It is based on Google's BERT algorithm, which pre-trains deep bidirectional representations by considering both left and right context in unlabeled text [4]. Other models like Mistral 7B – Instruct were engineered for superior performance and efficiency. One of the main achievements of the model is its ability to match the performance of other models that are three times larger in size. [13].

3 Data Set

3.1 Data Processing

We utilized 526 successive Federal Reserve Beige Books, spanning from May 1970 to December 2024 for our research. The complete historical database of Beige Books is sourced from the public archive maintained by the Federal Reserve Bank of Minneapolis and consists of 56,330 entries, covering 55 years of information distributed across 12 districts and around 10 economic topics. Each entry contains a descriptive text that explains the economic situation for a topic, at a given time, in a defined district.

3.2 Topic Labeling

Our objective is to measure economic sentiment at both the national and district levels with a focus on topics. A key challenge in analyzing the Beige Book is the variance in structure and content over time and between the national summary and District reports. While common topics such as "Summary of Economic Activity," "Labor Markets," and "Prices" are recurrent, the specific sections covered vary by District and have evolved. Notably, most of the Beige Books often lack clear topic labeling. To address this, we developed tailored data processing pipelines for each.

3.2.1 District Report Topic Labeling. To standardize these diverse reports for our topic-based sentiment analysis, we implemented a two-step process.

First, we defined a comprehensive list of key economic topics. This list was curated to capture the most frequent overlapping themes across all Beige Books and to align with the primary economic indicators that Federal Reserve officials typically monitor.

Second, we employed an LLM to categorize report content by topic, processing text in paragraph-long "chunks" rather than sentences. This approach was informed by the findings of Dunn et al. [5], who demonstrated that providing LLMs with larger segments of text, or "chunks," significantly improves their topic identification accuracy. They observed that this method allows the model to leverage a richer informational context, a principle that is particularly relevant to the anecdotal and comparative nature of the Beige Book. The narrative style often draws comparisons to previous periods, and these contextual cues are better captured at the paragraph level, leading to more precise topic labeling by Gemini 1.5 Flash.

3.2.2 National Summary Topic Labeling. Our analysis begins by harmonizing the structure of the national summary of the Beige Book reports across time. The current national summary typically follows a consistent structure, with dedicated sections for "Overall Economic Activity," "Inflation," and "Labor Markets," followed by highlights from each District. However, prior to 2017, national summaries were organized by economic sector (e.g., employment, manufacturing, consumer spending), while post-2017 reports shifted to a district-level narrative format. To ensure consistency, we restructured the district-level summaries from 2017 to 2024 by regrouping them into sectoral themes, effectively reconstructing a unified, topic-based format for the entire series.

4 Sentiment Analysis and Model Selection

4.1 District Report Sentiment Score

For the sentiment analysis phase, we evaluated the performance of five leading LLMs: GPT-4o, Claude 3.5, Gemini 1.5 Flash, FinBERT, and Mistral. The first three are closed, general-purpose language models, while the latter two are open-source models specialized for finance or sentiment analysis and can be further fine-tuned for specific tasks. By comparing both types, we can more rigorously assess the strengths and limitations of off-the-shelf versus customizable models, helping us identify which are best suited for nuanced financial sentiment analysis.

To create a robust benchmark, we randomly selected a representative paragraph from 10% of our 6,610 regional Beige Book reports, ensuring the sample was proportional to both region and topic. A human expert, such as a research analyst with a master's degree in economics and relevant industry experience, manually annotated this sample for sentiment on both a three-tier (-1, 0, 1) and a five-tier (-2,-1,0,1,2) scale.

All 660 paragraphs were processed by each of the five models GPT-4o, Claude 3.5, Gemini 1.5 Flash, FinBERT, and Mistral 7B

Instruct v0.2 for sentiment analysis. For the proprietary general-purpose models (GPT-4o, Claude 3.5, Gemini 1.5 Flash), we submitted the text directly via their respective APIs.

Our prompts were designed to optimize performance by incorporating three key strategies: 1) Role-playing: The prompt instructed the model to act as an economic researcher. 2) Few-shot learning: We provided a small number of pre-labeled examples to guide the model's responses. 3) Reasoning: We adopted a Label-then-Rationalize (LiRA) prompting structure, where the model is asked to first output a sentiment label and then briefly explain it. This follows Lopez-Lira and Tang [15], who show that reverse-CoT ("commit-then-explain") prompting improves consistency in financial text classification, and Vamvourellis and Mehta [21], who find that concise post-hoc reasoning often outperforms extended CoT in financial sentiment tasks. Collectively, these results suggest that for subjective financial interpretation, intuitive commitment followed by brief justification yields greater stability and interpretability than straightforward CoT("Let's think step by step") reasoning and no reasoning. 4) Date and time information is masked when possible to avoid look-ahead bias. This follows the caution from Glasserman and Lin [10], which shows that including future timestamps or sequence cues in sentiment input can leak information and inflate predictive performance.

Masked Model Input	Model Output (Sentiment + Rationale)
"Our respondents generally expected economic activity in the **second half of [YEAR]** to be higher for their states than for the nation as a whole. They were particularly optimistic about the prospects for consumer spending and plant and equipment investment. They also felt that inventory investment would continue strong."	`THREE_LEVEL: 1 FIVE_LEVEL: 1` Respondents "generally expected economic activity" to be higher in their states than nationwide, with particular optimism about consumer spending and plant/equipment investment. Inventory investment was also expected to "continue strong."
"**In [REGION]**, a discount department store to be opened in May processed 5,600 applications for 300 prospective jobs. A job conference for prospective college graduates in marketing was able to accept only 800 of 1,400 applicants for interviews, compared with a total of 350 applicants a year ago. The weakness of the job market reflects not only layoffs but hiring freezes and a sharply reduced rate of voluntary separations."	`THREE_LEVEL: -1 FIVE_LEVEL: -2` The section reveals significant job market weakness, with 5,600 applicants competing for just 300 jobs at one store. A job conference could only accept 800 of 1,400 applicants, compared to 350 total applicants the previous year. The weakness reflects "layoffs," "hiring freezes," and "sharply reduced rate of voluntary separations."
"Industry-wide consolidation has increased contacts' reliance on partnerships and subcontracts with other personnel supply firms. Vendor-on-premises operations dominate, hurting smaller local temp firms and favoring national companies large enough to meet the full range of employment demands of clients."	`THREE_LEVEL: 0 FIVE_LEVEL: 0` The section describes industry consolidation, increasing reliance on partnerships and subcontracts, with vendor-on-premises operations dominating. This hurts smaller local firms and favors national companies. The tone is matter-of-fact about structural changes rather than expressing clear sentiment.

Table 1: Illustrative examples of masked inputs and corresponding model outputs. The first example masks "1973," the second masks "Chicago," and the last remains unaltered.

To mitigate the effects of random variability in the models' outputs, we submitted each paragraph ten times to each LLM and selected the most frequently occurring sentiment score. In cases of a tie (e.g., five positive and five neutral ratings), the paragraph was re-evaluated, and the model's reasoning was manually reviewed to ensure its coherence and accuracy.

For FinBERT, we used the open-source model as-is, without further customization. The content section of each entry in the dataset served as the input and was limited to a maximum of 512 tokens due to technical constraints of the FinBERT model and 768 tokens for Mistral. These tokens were passed through the models to obtain a probability distribution over three possible sentiment labels: 1 = Positive, 0 = Neutral, and -1 = Negative.

Mistral 7B Instruct v0.2, also open-source, was applied in its base instruction-following form, without additional training on financial data. This approach allowed us to compare model performance across a range of architectures and specialization levels, using a consistent set of input paragraphs.

4.2 District Report Model Selection

Our benchmarking process revealed that the LLMs can achieve a high degree of accuracy in sentiment classification, with results that are comparable to those of a human expert: Among the five models evaluated, the models generated more comparable results on 3-tier sentiment labeling than 5-tier, indicating the ability to identify directional change but with limited granularity.

Specifically, with 3-tier labeling, GPT-4o and Claude 3.5 consistently deliver the highest sentiment accuracy across all topics, with averages of 85.9% and 80.6%, respectively. Gemini 1.5 Flash (72.0%) and Mistral (70.6%) perform lower but still achieve robust results across most sections. FinBERT (44.5%) underperformed, showing low accuracy in nearly every category.

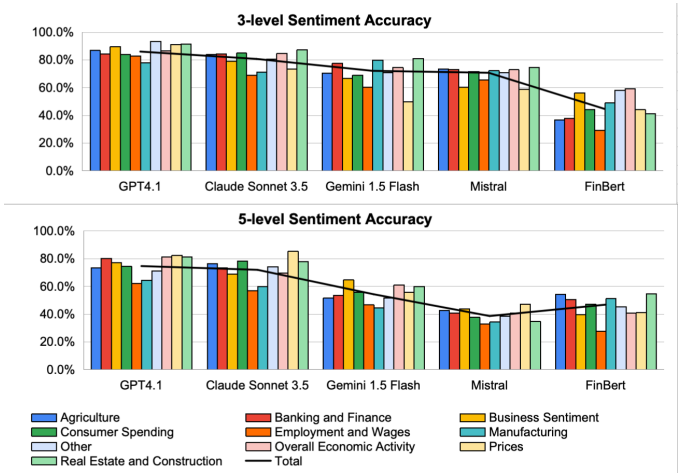


Figure 1: Accuracy of 3- and 5-Level Sentiment Models

This result is unsurprising given the linguistic characteristics of institutional texts. The Beige Book, FOMC minutes, and corporate filings employ cautious, qualified language that eschews emotional extremes. Fine-grained sentiment distinctions thus add complexity without informational gain, forcing more paragraphs into weakly differentiated "mildly positive/negative" categories, thereby reducing the accuracy of our sentiment signals.

Given these findings, we decided to use 3-tier sentiment and selected Gemini 1.5 Flash as our primary closed-source model due to its solid accuracy, free accessibility, and practical balance of performance and cost. We also kept the two open-source models, FinBERT and Mistral, which allowed us to benchmark proprietary models against open alternatives since Gemini and Mistral shared similar sentiment accuracy, and to test whether sentiment labeling accuracy has a meaningful impact on our logistic regression models predicting recession risk, as FinBERT has a lower accuracy.

4.3 National Summary Sentiment Score

For extracting the sentiments from the national summary,we employed two main strategies: FinBERT and the Loughran–McDonald (LM) Dictionary.

4.3.1 FinBERT Sentiment of National Summary. To extract sentiment from the national summary entry, we started with FinBERT. Just as we have extracted sentiment from the full Beige Book sections, FinBERT here outputs probabilistic scores for positive, neutral, and negative classes. We extracted the sentiment label by selecting the class with the highest predicted probability, then mapped each label to a numeric value in three classes: +1 (positive), -1 (negative), or 0 (neutral). The final monthly sentiment score was computed by averaging the sentiment values across all paragraphs. FinBERT scoring was implemented using the Hugging Face transformers library [22], with each paragraph tokenized and passed through the model, followed by mean-pooling of sentiment outputs. The sentiment scoring was applied with a progress-tracked loop to handle the full harmonized national summary dataset from 1970 to 2024.

4.3.2 Loughran-McDonald (LM) Dictionary. In addition to the FinBERT transformation, we also implemented a traditional dictionary-based sentiment scoring approach using the Loughran-McDonald (LM) Master Dictionary, which was developed specifically for financial text to avoid misclassifications [16]. After preprocessing the LM lexicon and national summary text (lowercasing, tokenization, and word frequency counting), we computed paragraph-level sentiment as the difference between counts of LM-defined positive and negative words. Each summary paragraph was assigned a sentiment value out of three classes: +1 if positive word counts exceeded negative, -1 if negative exceeded positive, and 0 otherwise. As with FinBERT, monthly sentiment scores were derived by averaging across all paragraphs.

5 Recession Forecasting with Beige Book Sentiment

5.1 District Report Sentiment

5.1.1 Data Preparation. First, following the Cleveland Fed methodology [6], we measured topic-level tone as the difference between positive and negative paragraphs divided by their total, capturing the cross-regional balance of positive and negative sentiment in each report. To build on this, we computed a simple mean of raw sentiment scores across regions for each topic, providing a richer measure that captures both the degree and intensity of positivity or negativity across regions. Lastly, since recessions are typically characterized by rapidly deteriorating conditions, we calculated month-over-month changes in sentiment in order to identify significant and fast-moving declines that could serve as an early recession signal.

5.1.2 Logistic Regression Baseline. We estimate h-month-ahead predictive (binary) logistic regressions of US recessions at horizons ranging from h = -6/-3 months (backcast), h = 0 (nowcast), to h = +3/+6 months (forecast). To construct the sentiment features, we generate topic-level sentiment scores using three different language models (Gemini 1.5-Flash, FinBERT, and Mistral) and process these raw scores using the three abovementioned alternative calculation methods. The resulting sentiment indices are then used as predictors in the logistic regression framework to forecast the binary outcome of whether a given observation signals a recession (1) or not (0). We trained Complete (C) models using all independent variables, regardless of their statistical significance, and Reduced (R) models by removing those with p-values greater than 0.05.

This approach allows us to empirically test whether our sentiment indices serve as leading, coincident, or lagging indicators of the U.S. business cycle, as reflected in these in-sample relationships. Given the significant class imbalance in our dataset (with recession periods comprising only about 10% of observations), we evaluate model performance primarily using the F1 score for the positive (recession) class and balanced accuracy.

In-sample Results. Disaggregated topic-level sentiment from the Beige Book provides powerful real-time signals of the U.S. business cycle, with performance peaking at the nowcast horizon (h = 0). Logistic regressions using simple sentiment averages achieve F1 scores up to 0.89 and balanced accuracy near 0.98 (Mistral-based). The Mistral-based model also achieves near-perfect and stable AUCs (0.85–1.00) across horizons, confirming its robustness and generalization. The Gemini-based model achieved similar competitive results with an F1 score of 0.96 and a balanced accuracy of 0.89, but its performance decays faster in both forecasting and backcasting. FinBERT maintains consistently high AUCs, particularly in backcast horizons (AUC>0.9), suggesting stronger sensitivity to early signs of financial stress preceding recessions.

Beyond nowcasting, topic-level sentiment continues to exhibit non-trivial predictive power at short leads and lags. For the Gemini-based model, pseudo R^2 peaks at 0.67 (h=0) and remains sizable at 0.48 for a three-month lead (h=+3), highlighting that sentiment contains meaningful forward-looking information. Consistent patterns appear across models: Mistral and FinBERT achieve F1 scores around 0.7 and balanced accuracy near 0.9 at h=+3. At longer forecast horizons, predictive power generally declines, consistent with the idea that sentiment is most informative about contemporaneous economic conditions. These findings highlight the Beige Book's value as a near-term leading indicator, motivating our subsequent out-of-sample analysis to assess the robustness and real-time applicability of this predictive relationship.

Lead–Lag	Precision(1)	Recall(1)	F1-Score	BA	AUC	pseudo R^2
−6 months	0.60	0.33	0.86	0.65	0.88	0.30
−3 months	0.78	0.62	0.92	0.79	0.95	0.49
0 months	0.86	0.80	0.96	0.89	0.97	0.67
+3 months	0.69	0.53	0.91	0.75	0.95	0.48
+6 months	0.42	0.18	0.85	0.57	0.87	0.28

Table 2: Logistic regression results using Gemini sentiment (in-sample). Precision and recall refer to the recession class (label = 1).

5.1.3 Out-of-Sample Logistic Regression with Expanding-Window. We complement the in-sample evidence with out-of-sample "real-time" prediction based on a recursive expanding window. At every month t, we re-estimate the model on an expanding window that includes data up to t-1, adding observations one at a time, as to reproduce the information available at t; for h>0 we additionally trim the most recent h months from the estimation window so that

Model	Lead-Lag	F1 (C)	F1 (R)	BA (C)	BA (R)	AUC (C)	AUC (R)	Thres. (C)	Thres. (R)
Mistral	-6 mo	0.74	0.78	0.86	0.88	0.85	0.86	0.15	0.17
	-3 mo	0.85	0.76	0.92	0.93	0.95	0.97	0.21	0.04
	0 mo	0.89	0.78	0.98	0.93	1.00	0.97	0.37	0.35
	+3 mo	0.67	0.68	0.91	0.92	0.95	0.94	0.11	0.11
	+6 mo	0.44	0.56	0.82	0.86	0.88	0.86	0.07	0.15
FinBERT	-6 mo	0.62	0.59	0.84	0.81	0.83	0.79	0.17	0.17
	-3 mo	0.76	0.62	0.87	0.87	0.94	0.92	0.38	0.12
	0 mo	0.69	0.62	0.93	0.90	0.94	0.90	0.32	0.25
	+3 mo	0.73	0.69	0.91	0.89	0.92	0.92	0.29	0.24
	+6 mo	0.60	0.43	0.84	0.82	0.88	0.85	0.21	0.09

Table 3: Logistic regression results using Mistral and FinBERT sentiment(in-sample).

every training label $y_{i,h}$ is dated no later than the end of the window. We initialize with at least 120 months of data, apply a one-month embargo between the last training observation and the test point, and score X_t to obtain the h-month recession probability $\hat{p}_{t,h}$.

Based on the work of Gascon and Martorana [8], we construct a rule that we think a recession signal occurs immediately if the model's probability for a particular month p_t is very high (at least 90%). Otherwise, we look back over the most recent four Beige Books: if at least three of those months had probabilities above 85%, then the current month is also classified as a recession. If neither of these conditions is met, the period is classified as an expansion.

For example, suppose that three out of the last four Beige Books each had predicted probabilities above 85%. Even if the most recent probability is below 90%, the economy would still be considered to be in recession under this rule. Conversely, if only one or two months in the past four exceed the 85% threshold, the expansion state is maintained.

$$\hat{r}_t^{\text{raw}} = \begin{cases} 1, & \text{if } p_t \geq 0.85, \\ 0, & \text{if } p_t < 0.85. \end{cases} \tag{1}$$

$$\hat{R}_t = \begin{cases} 1, & \text{if } p_t \geq 0.90, \\ 1, & \text{if } \sum_{s=t-m+1}^{t} \hat{r}_s^{\text{raw}} \geq k, \quad \text{with } k = 3,\ m = 4, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

5.1.4 Out-of-sample Results. The out-of-sample validation confirms the robustness of our in-sample findings. Using an expanding-window approach, all three models maintain strong nowcasting performance (F1 scores: Gemini 0.85, Mistral 0.80, FinBERT 0.67), with only modest degradation from in-sample results. Beyond this point, both lead and lag performance generally deteriorate, consistent with information decay in sentiment signals over time.

For the Gemini-based model, predictive accuracy peaks at $h = 0$, with an F1-score of 0.85 and balanced accuracy of 0.93, indicating that the model effectively captures real-time cyclical conditions using only information available at the time of estimation. Performance remains solid at $h = -3$ (F1=0.68, BA=0.80), suggesting that sentiment still contains valuable lagged signals of economic turning points. Predictive power, however, decays markedly at $h = +3$ and beyond, with F1-scores falling below 0.45 and balanced accuracy around 0.69.

Similarly, the Mistral-based model also delivers consistently strong out-of-sample performance, sustaining an AUC of 0.95 and balanced accuracy above 0.90 at the nowcast horizon ($h = 0$). Notably, the Mistral-based model also shows less deterioration across both leading and lagging horizons, indicating more persistent capture of underlying cyclical patterns.

All three models deliver broadly consistent OOS patterns (Table 4) as they all maintain balanced accuracy above 0.9 at nowcast horizons, recall above 0.85, and AUC remains in the 0.85 to 0.95 range. These consistently robust results across models with different knowledge cutoffs support the idea that Beige Book sentiment generalizes well out of sample, providing reliable real-time information about the business cycle, particularly for nowcasting.

Model	Lead–Lag	Precision(1)	Recall(1)	F1-Score	BA	AUC
Gemini	−6 mo	0.53	0.22	0.31	0.60	0.79
	−3 mo	0.74	0.62	0.68	0.80	0.92
	0 mo	0.80	0.89	0.85	0.93	0.96
	+3 mo	0.42	0.46	0.44	0.69	0.89
	+6 mo	0.20	0.19	0.20	0.56	0.79
Mistral	−6 mo	0.59	0.83	0.69	0.89	0.90
	−3 mo	0.45	0.87	0.60	0.89	0.95
	0 mo	0.76	0.85	0.80	0.91	0.95
	+3 mo	0.41	0.70	0.52	0.81	0.94
	+6 mo	0.19	0.52	0.28	0.67	0.85
FinBERT	−6 mo	0.57	0.83	0.68	0.89	0.87
	−3 mo	0.51	0.78	0.62	0.86	0.94
	0 mo	0.53	0.88	0.67	0.91	0.94
	+3 mo	0.50	0.83	0.62	0.88	0.94
	+6 mo	0.29	0.65	0.41	0.76	0.86

Table 4: Out-of-sample logistic regression results by LLM-based sentiment model. Precision and recall refer to the recession class (label = 1).

5.1.5 Nowcasting with XGBoost using Embeddings. Building on the logistic regression benchmarks, we next examine whether richer textual representations can enhance predictive performance. Unlike scalar sentiment indices, embeddings capture nuanced semantic information embedded in the Beige Book narrative. To this end, we use the FinBERT model to generate dense paragraph-level representations: each paragraph within a section is tokenized and passed through FinBERT, from which we extract the hidden states of the final transformer layer and compute a mean pooling across all tokens, yielding a 768-dimensional embedding vector for each topic paragraph.

We consider two modeling strategies using the FinBERT embeddings. In the Independent Section Classifier setup, each section's embedding is fed into a separate XGBoost model to isolate its individual predictive contribution. In the Aggregated Embedding Classifier, all section embeddings were concatenated into a single vector for a unified XGBoost model, enabling the capture of cross-section interactions. Consistent with our logistic regression approach, models were evaluated using precision, recall, F1-score, and ROC-AUC, focusing on recession (class 1) detection.

Results and Discussion. The aggregated model achieved 90.5% overall accuracy with strong expansion detection (precision 0.95, recall 0.94, F1 0.95) but weaker recession identification (precision 0.64, recall 0.69, F1 0.67). Feature importance analysis revealed heavy reliance on "Overall Economic Activity" (54%+ of feature gain), followed by "Real Estate and Manufacturing". In contrast, single-topic models showed variable performance. "Banking and Finance" and "Overall Economic Activity" achieved recession F1-scores of 0.69–0.72, while "Employment", "Prices", and "Agriculture"

performed poorly (recall <0.25, F1 0.1–0.3), indicating limited standalone predictive capacity.

5.1.6 Discussion of District-Level Findings. **Topic Signals and Macroeconomic Intuition.** A key advantage of our Beige Book based approach is that the models are inherently explanatory and economically intuitive. Unlike traditional recession indicators, which often provide limited insight into the underlying drivers of the predicted probability, our topic-level sentiment framework allows us to trace which aspects of the economy contribute to rising or weakening recession risks. Our in-sample analysis reveals consistent patterns in how topic-level sentiment contributes to recession prediction across models and forecast horizons.

Gemini-based models show that sentiment in *"Employment and Wages"*, *"Manufacturing"*, and *"Real Estate and Construction"* is negative and highly significant for nowcasting, explaining 76% of model variance. Their predictive importance declines at longer leads, while *"Consumer Spending"* becomes dominant at h=+3, confirming its leading properties. Conversely, *"Employment and Wages"* remain influential in backcasting, consistent with the labor market's lagging behavior in the business cycle. For FinBERT sentiment, *"Overall Economic Activity"* is statistically significant for both nowcasting and backcasting, while *"Real Estate and Construction"* retains predictive power even at h=+6. *"Banking and Finance"* peaks around turning points (h=-3 to +3) but weakens beyond those horizons. For Mistral-based models, *"Real Estate and Construction"* dominates across horizons. *"Manufacturing"* is most effective for near-term prediction, and *"Banking and Finance"* and *"Agriculture"* are more relevant for backcasting.

Across all models, *"Real Estate and Construction"*, *"Employment and Wages"*, and *"Manufacturing"* consistently emerge as top predictors, particularly for nowcasting. These results demonstrate that topic-based sentiment analysis offers nuanced and timely insights into the drivers of the U.S. business cycle, with both model-specific and broadly consistent signals across time horizons.

Language Model Comparison. Across all three LLMs, in-sample logistic regression models using the simple average sentiment method consistently achieve the highest F1 score and balanced accuracy regardless of the horizon. None of the top models required class reweighting despite a severe imbalance, indicating that topic-level sentiment carries substantial information for recession detection.

Gemini performs best at nowcasting, with both precision and recall above 0.80 in-sample and out-of-sample. Unlike the other two models, Gemini's performance does not deteriorate when moving from in-sample to out-of-sample. However, compared with Mistral and FinBERT, its predictive strength decays more quickly in the lead–lag tests, particularly at the forecasting horizons, where it underperforms in both samples. This pattern suggests that Gemini's sentiment most accurately reflects contemporaneous economic conditions, but carries limited forward-looking information. It also highlights a limitation in our setup: Gemini is trained on a larger and more recent corpus than the other two models, so its stronger out-of-sample performance may partially reflect potential look-ahead bias in LLMs' general knowledge rather than purely superior generalization.

Mistral offers the most balanced and generalizable performance. In-sample BA remains above 0.9 across horizons, and AUC values approach 1.0. It outperforms FinBERT in nowcasting, achieving a balanced accuracy of 0.98 versus 0.93 in the full specification, and 0.93 versus 0.90 in the reduced models, with consistently higher F1 scores across horizons. Out-of-sample, it sustains strong recall ($\approx$ 0.83 at -6 months) and F1 ($\approx$ 0.69), showing resilience to horizon shifts.

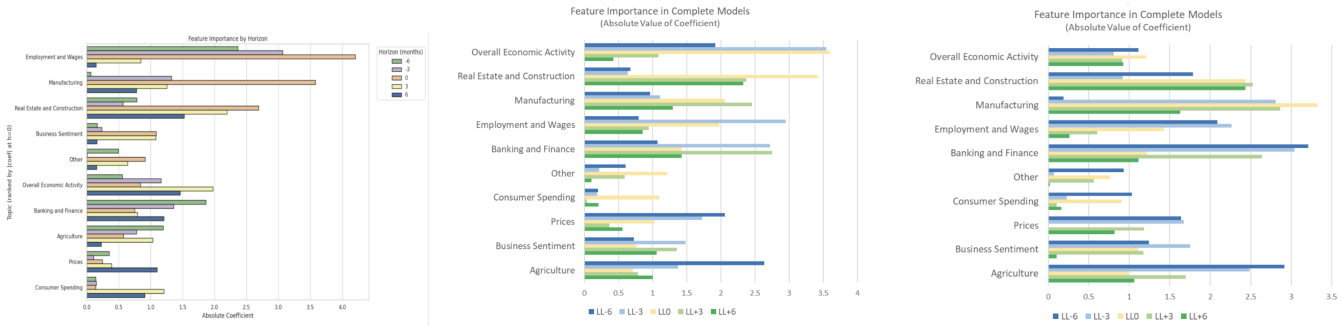
FinBERT maintains strong Balanced Accuracy and AUC from LL -6 to LL +6 both in-sample and out-of-sample, which shows that the model distinguishes well between recession and non-recession periods and generalizes robustly across horizons. FinBERT also shows stronger backcast ability out-of-sample at h=-6 and h=-3, with BA stable between 0.86 and 0.89 and F1 rising from 0.62 to 0.68 as the horizon moves backward. This may suggest that FinBERT's domain-specific financial sentiment is particularly sensitive to early deterioration in business tone and credit conditions preceding recessions, allowing it to identify recessionary patterns before they are reflected in broader economic narratives.

Compare Logit vs XGBoost. Despite capturing richer semantic information theoretically, in our experiments, FinBERT embeddings did not outperform simpler sentiment-based logistic regression. As Gentzkow et al. [9] note, simple interpretable features often prove more robust in economic prediction when sample sizes are limited. Our results support this: the logistic approach's superior out-of-sample performance may reflect either (i) greater informativeness of sentiment polarity versus dense representations, or (ii) the dilution of recession signals in high-dimensional semantic spaces. While XGBoost's regularization and implicit feature selection typically handle high-dimensional embeddings well, our results suggest that FinBERT's general-purpose representations may spread recession-relevant information thinly across 768 dimensions. In contrast, task-specific sentiment scores concentrate predictive signals into lower-dimensional, directly interpretable features, an advantage that proves decisive in our relatively small sample size.

Compare LLM-based Sentiment vs Traditional Recession Indicators. To better understand the predictive performance of our LLM-based Beige Book sentiment models, we compare their out-of-sample AUC values to those reported in Pike and Vazquez-Grande [17], who evaluate probit models' performance in recession in the next 12 months. In their study, a bivariate model combining yield curve spreads and corporate bond spreads achieves the highest OOS forecasting power, with an out-of-sample AUROC of 0.88. By comparison, our Mistral and Gemini sentiment models reach AUCs of 0.95–0.96 at $h = 0$ and remain at 0.94 at three months ahead and 0.85 at six months ahead in out-of-sample tests, indicating comparable or stronger predictive power than traditional yield curve based indicators.

5.2 National Summary Sentiment

Following the same approach used for the district-level analysis, we applied supervised learning to the national Beige Book summaries. Each monthly report was aggregated into a single sentiment score using FinBERT and other LLM-based measures. We implemented



(a) Gemini (b) FinBERT (c) Mistral

Figure 2: Feature importance across models (absolute coefficients).

and compared three classification models, including Logistic Regression (baseline), Random Forest, and Gradient Boosting Machine (GBM).

5.2.1 Model performance using FinBERT sentiment. We began our modeling with a logistic regression using only the FinBERT sentiment score. The model showed low precision (0.24), though accuracy for non-recession periods remained high. To address this limitation of linear models, we transitioned to Random Forest and Gradient Boosting models, as these nonlinear ensemble methods offer improved balance between precision and recall by effectively modeling more intricate feature interactions.

We estimate h-month-ahead predictive models of U.S. recessions at horizons $h = -6, -3$ (backcast), $h = 0$ (nowcast), and $h = +3, +6$ (forecast), using lagged average sentiment scores as predictors. As shown, model performance remained broadly consistent across classifiers and forecast horizons (Table 5).

Horizon	Model	Precision	Recall	F1-Score	Accuracy
Panel A: LM Sentiment (In-Sample)					
h=0	Logistic Regression	0.19	0.95	0.31	0.51
h=0	Random Forest	0.18	0.76	0.29	0.57
h=0	Gradient Boosting (GBM)	0.20	0.89	0.33	0.57
h=3	Logistic Regression	0.17	0.84	0.28	0.49
h=3	Random Forest	0.16	0.62	0.25	0.58
h=3	Gradient Boosting (GBM)	0.19	0.84	0.31	0.57
h=6	Logistic Regression	0.17	0.87	0.29	0.50
h=6	Random Forest	0.17	0.69	0.27	0.56
h=6	Gradient Boosting (GBM)	0.18	0.76	0.29	0.56
Panel B: FinBERT Sentiment (In-Sample)					
h=0	Gradient Boosting (GBM)	0.27	0.89	0.41	0.71
h=0	Logistic Regression	0.24	0.91	0.38	0.65
h=0	Random Forest	0.26	0.80	0.40	0.71
h=3	Gradient Boosting (GBM)	0.25	0.82	0.38	0.69
h=3	Logistic Regression	0.22	0.82	0.34	0.63
h=3	Random Forest	0.25	0.75	0.37	0.71
h=6	Gradient Boosting (GBM)	0.26	0.84	0.40	0.71
h=6	Logistic Regression	0.23	0.89	0.37	0.64
h=6	Random Forest	0.24	0.75	0.37	0.70

Table 5: In-sample performance of LM- and FinBERT-based models across forecast horizons.

Next, we analyzed which Beige Book sections most influenced recession classifications and how their importance varied across horizons. As shown in Figure 3, "Employment", "Financial Conditions", and "Finance and Credit" consistently ranked as top predictors at all horizons ($h = 0, 3, 6$), underscoring the critical role of topic-specific sentiment in signaling recessionary conditions.

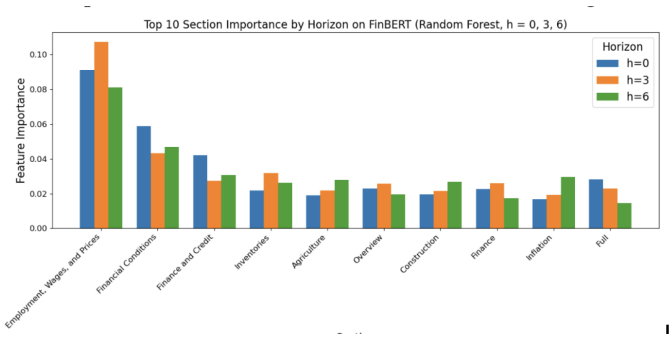


Figure 3: Top 10 predictors of US recession from Random Forest model using FinBERT sentiment using various time horizons

5.2.2 Model performance using LM sentiment. A similar modeling approach was applied using the LM (Loughran-McDonald) sentiment scores. As shown in Table 5, logistic regression again demonstrated high recall but very low precision, highlighting its tendency to over-predict recessions while accurately identifying non-recession periods. Random Forest and Gradient Boosting models outperformed logistic regression by providing a more balanced trade-off between precision and recall across all forecast horizons (h = 0, 3, and 6). Notably, Gradient Boosting yielded the highest F1-scores across all horizons, suggesting its superior ability to capture the nonlinear relationships present in the LM sentiment data.

Replicating the same analysis at the topic level using LM sentiment, we identified the Beige Book sections most influential in classifying recessions. As shown in Figure 4, "Financial Conditions", "Finance", "Prices", "Wages", and "Inflation" consistently ranked as top predictors across all horizons ($h = 0, 3, 6$), underscoring the importance of section-specific narratives for improving recession forecasting using LM sentiment.

5.2.3 Analysis of national summary-based Models. Analysis of sentiment derived from both FinBERT and Loughran-McDonald (LM) lexicons on the national summary revealed consistently low precision across all forecasting horizons and models for predicting recessions, whereas accuracy for non-recession periods remained comparatively high (Table 5). Although the national summary is an extreme distillation of the Beige Book and may not capture all

nuanced signals, qualitative assessments showed that predicted recession patterns aligned reasonably well with actual NBER-defined recession episodes. Notably, the top predictors identified by both sentiment approaches exhibited considerable overlap, emphasizing key economic themes such as employment, financial conditions, prices, wages, and inflation.

It is worth noting that the FinBERT sentiment distribution was somewhat imbalanced, tending toward extreme positive or negative values with fewer neutral scores, which may have influenced model calibration. Despite applying class weighting, threshold adjustments, and ensemble methods to address imbalance, the models struggled to significantly improve recession case precision, indicating the inherent challenges in forecasting rare but critical downturn events. These findings suggest that future research should explore additional data sources, feature engineering, or hybrid modeling strategies to better capture early recession signals and enhance predictive accuracy.

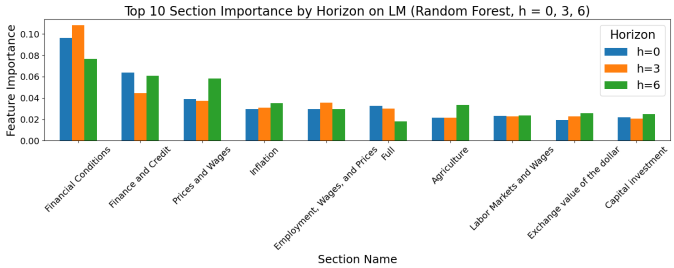


Figure 4: Top 10 predictors of US recession from Random Forest model using LM sentiment using various time horizons.

5.2.4 Analysis of National Summary OOS models. We applied the same expanding-window out-of-sample framework used for the district-level models on the national summary data. The LM sentiment model demonstrates modest but stable generalization from in-sample to out-of-sample periods. Recall remains consistently near the 0.80 target threshold across forecast horizons, and precision experiences minimal deterioration across classifiers. Notably, optimal LM thresholds for Logistic Regression were moderate (primary 0.60–0.65, secondary 0.80–0.85, k=3, m=4–5), indicating a weaker underlying signal requiring persistence rules to filter spurious single-month spikes. FinBERT sentiment demonstrates substantially stronger out-of-sample performance, particularly for Logistic Regression. GBM remains feasible across horizons but experiences precision decay OOS. The optimal threshold configuration for FinBERT—primary thresholds of 0.85–0.90, secondary 0.70, with persistence parameters k=3, m=4–5, reflecting a sharper, more concentrated signal distribution. The higher primary threshold indicates greater separability between recession and non-recession regimes, requiring stricter trigger criteria to optimize precision under the binding recall constraint.

6 Recession Probability Indicator

Using the logistic models we built with district report sentiment by different LLM models, we are able to plot real-time conditional probability of US recession during the Beige Book reporting period.

Horizon	Model	Precision	Recall	F1-Score	Accuracy
Panel A: LM Sentiment (Out-of-Sample)					
h=0	Gradient Boosting (GBM)	0.17	0.81	0.28	0.54
h=0	Logistic Regression	0.18	0.86	0.30	0.55
h=0	Random Forest	0.17	0.74	0.27	0.56
h=3	Gradient Boosting (GBM)	0.17	0.80	0.28	0.54
h=3	Logistic Regression	0.19	0.80	0.30	0.58
h=3	Random Forest	0.17	0.70	0.27	0.59
h=6	Gradient Boosting (GBM)	0.17	0.78	0.28	0.55
h=6	Logistic Regression	0.17	0.85	0.28	0.51
h=6	Random Forest	0.17	0.68	0.27	0.60
Panel B: FinBERT Sentiment (Out-of-Sample)					
h=0	Gradient Boosting (GBM)	0.23	0.80	0.35	0.67
h=0	Logistic Regression	0.33	0.80	0.46	0.79
h=0	Random Forest	0.21	0.74	0.33	0.66
h=3	Gradient Boosting (GBM)	0.22	0.80	0.35	0.67
h=3	Logistic Regression	0.29	0.82	0.42	0.75
h=3	Random Forest	0.20	0.72	0.32	0.65
h=6	Gradient Boosting (GBM)	0.22	0.80	0.34	0.65
h=6	Logistic Regression	0.21	0.81	0.34	0.64
h=6	Random Forest	0.21	0.73	0.32	0.66

Table 6: Out-of-sample performance of LM- and FinBERT-based models across horizons.

These model-based probabilities match well with the recession periods defined by NBER. Higher probabilities in the three graphs of Figure 5 are associated with NBER-defined recessions, but different models displayed different levels of sensitivity. As seen in the top panel, Gemini produces cleaner, more conservative signals, while Mistral is somewhat more sensitive and "reactive", as seen in the middle, which may be useful for early warning. In the bottom panel of Figure 5, FinBERT also showed high correlation with actual recession periods, but suffered from higher false positives and appeared noisier visually.

For national-summary-based sentiment, we plotted the predicted probabilities of recession against the actual NBER-defined recession periods using a six-month forecasting horizon (h = 6), using FinBERT-based (top) and LM-based (bottom) sentiment models. As shown in Figure 6, the signals from both models align closely with actual recessions, with all three models—Logistic Regression, Random Forest, and Gradient Boosting—successfully capturing major downturns, including the early 1980s, early 2000s, and the 2008 global financial crisis. Both models also indicate a rising probability of recession late 2024, suggesting that sentiment-based forecasts may provide valuable early warning signals for future downturns.

7 Conclusion

Our research shows that large language models can extract useful topic-level sentiment from the Beige Book for real-time monitoring and short-term prediction of U.S. recessions. The models generalize well out of sample, sustaining AUCs above 0.9 and balanced accuracy near 0.8, comparable or better than traditional yield curve-based recession predictors. Topic-based sentiment from district reports yields the highest predictive accuracy, and sentiment in "Employment and Wages", "Manufacturing", and "Real Estate and Construction" consistently emerged as leading indicators across models. In comparison, national summary sentiment measured with both FinBERT and Loughran-McDonald dictionaries also aligns well with NBER-defined recessions, especially when modeled with nonlinear classifiers. Key sections such as employment, financial conditions, and prices emerge as important predictors even in this aggregate national view, supporting the broader robustness of sentiment-based recession indicators. Together, these findings highlight the value of

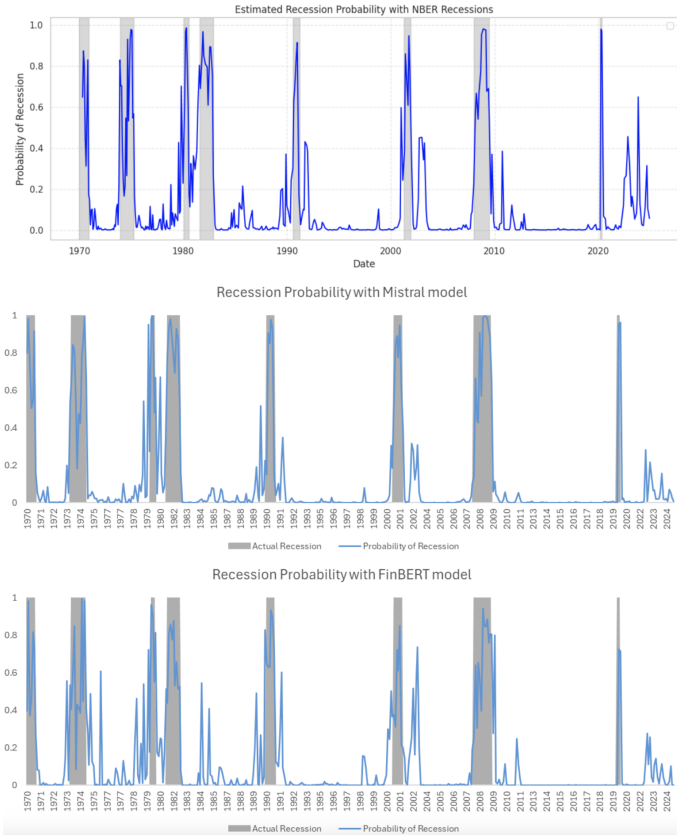


Figure 5: Recession probability from district sentiment (Gemini, Mistral, and FinBERT)

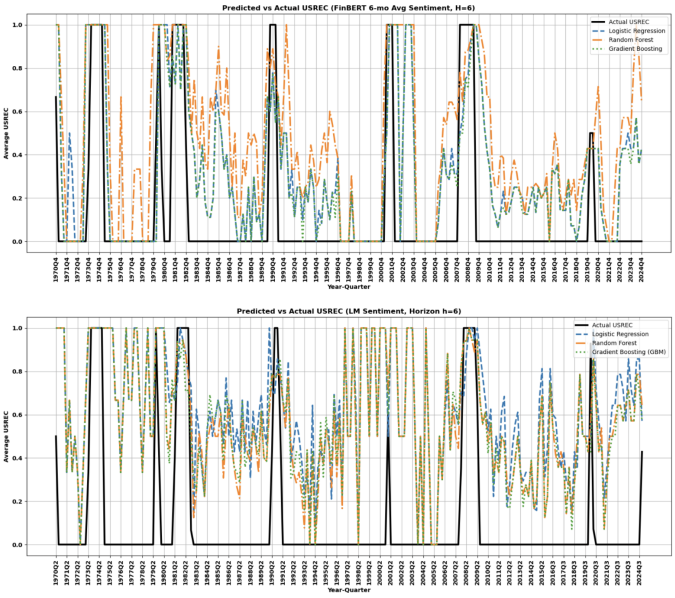


Figure 6: Recession probability from national summary sentiment (FinBERT and LM)

topic-level sentiment analysis using LLMs on Beige Book textual data for timely macroeconomic risk assessment.

References

[1] Michelle T. Armesto et al. 2009. Measuring the information content of the beige book: A mixed data sampling approach. 35–55 pages.

[2] Moeko Asano et al. 2024. Analysis of investment behavior of individual investors in the FX market: Influence of FOMC and Beige Book information.

[3] Nathan S. Balke, Michael Fulmer, and Ren Zhang. 2017. Incorporating the beige book into a quantitative index of economic activity. 497–514 pages.

[4] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding.

[5] Wendy Dunn, Ellen E. Meade, Nitish Ranjan Sinha, and Raakin Kabir. 2024. Using Generative AI Models to Understand FOMC Monetary Policy Discussions. FEDS Notes. Washington: Board of Governors of the Federal Reserve System. Available at https://doi.org/10.17016/2380-7172.3678.

[6] Ilias Filippou, Christian Garciga, James Mitchell, and My T. Nguyen. 2024. Regional Economic Sentiment: Constructing Quantitative Estimates from the Beige Book and Testing Their Ability to Forecast Recessions. *Federal Reserve Bank of Cleveland, Economic Commentary* 2024-08 (2024). doi:10.26509/frbc-ec-202408

[7] Masaki Fujiwara, Yoshiyuki Suimon, and Kei Nakagawa. 2023. Treasury yield spread prediction with sentiments of Beige Book and macroeconomic data.

[8] Charles S. Gascon and Joseph Martorana. 2024. *The Beige Book and the Business Cycle: Using Beige Book Anecdotes to Construct Recession Probabilities.* Technical Report Working Paper 2024-037. Federal Reserve Bank of St. Louis. https://doi.org/10.20955/wp.2024.037

[9] Matthew Gentzkow, Bryan Kelly, and Matt Taddy. 2019. Text as Data. *Journal of Economic Literature* 57, 3 (2019), 535–574.

[10] Paul Glasserman and Caden Lin. 2023. Assessing Look-Ahead Bias in Stock Return Predictions Generated By GPT Sentiment Analysis. *arXiv preprint arXiv:2309.17322* (2023). arXiv:2309.17322 [q-fin.GN] doi:10.48550/arXiv.2309.17322

[11] Anne Lundgaard Hansen and Sophia Kazinnik. 2024. Can chatgpt decipher fedspeak? Available at SSRN.

[12] Allen H. Huang, Hui Wang, and Yi Yang. 2023. FinBERT: A large language model for extracting information from financial text. 806–841 pages.

[13] Albert Qiaochu Jiang et al. 2023. Mistral 7B. ArXiv abs/2310.06825.

[14] Alex Kim, Maximilian Muhn, and Valeri Nikolaev. 2024. Financial statement analysis with large language models. Fama-Miller working paper. Available at SSRN: https://ssrn.com/abstract=4835311.

[15] Alejandro Lopez-Lira and Yuehua Tang. 2023. Can chatgpt forecast stock price movements? return predictability and large language models. *arXiv preprint arXiv:2304.07619* (2023).

[16] Tim Loughran and Bill McDonald. 2011. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. 35–65 pages.

[17] Tyler Pike and Francisco Vazquez-Grande. 2019. *Out-of-Sample Performance of Recession Probability Models.* FEDS Notes. Board of Governors of the Federal Reserve System. doi:10.17016/2380-7172.2454 Washington: Board of Governors of the Federal Reserve System, December 13, 2019.

[18] Bryan R. Routledge. 2019. Machine learning and asset allocation. 1069–1094 pages.

[19] Shibley Sadique et al. 2013. Soft information and economic activity: Evidence from the Beige Book. 81–92 pages.

[20] Bennett Saltzman and Julieta Yung. 2018. A machine learning approach to identifying different types of uncertainty. 58–62 pages.

[21] Dimitris Vamvourellis and Dhagash Mehta. 2025. Reasoning or Overthinking: Evaluating Large Language Models on Financial Sentiment Analysis. *arXiv preprint arXiv:2506.04574v1* (2025). https://arxiv.org/abs/2506.04574v1 BlackRock, Inc..

[22] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, and Tim Rault. 2020. HuggingFace's Transformers: State-of-the-art Natural Language Processing. 38–45 pages.

[23] Kim Wonseong, Jan Frederic Spörer, and Siegfried Handschuh. 2023. Analyzing fomc minutes: Accuracy and constraints of language models. Affiliated in Korea University.